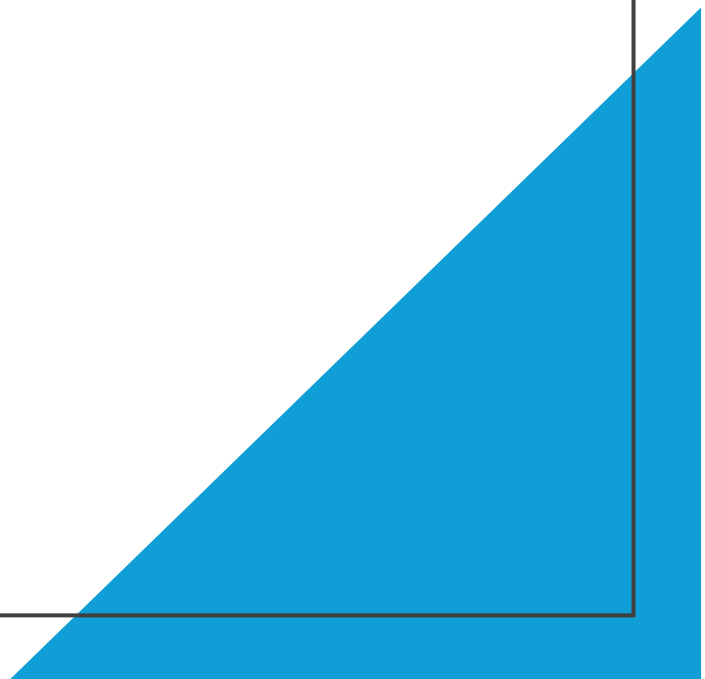


DS1 PROJEKT

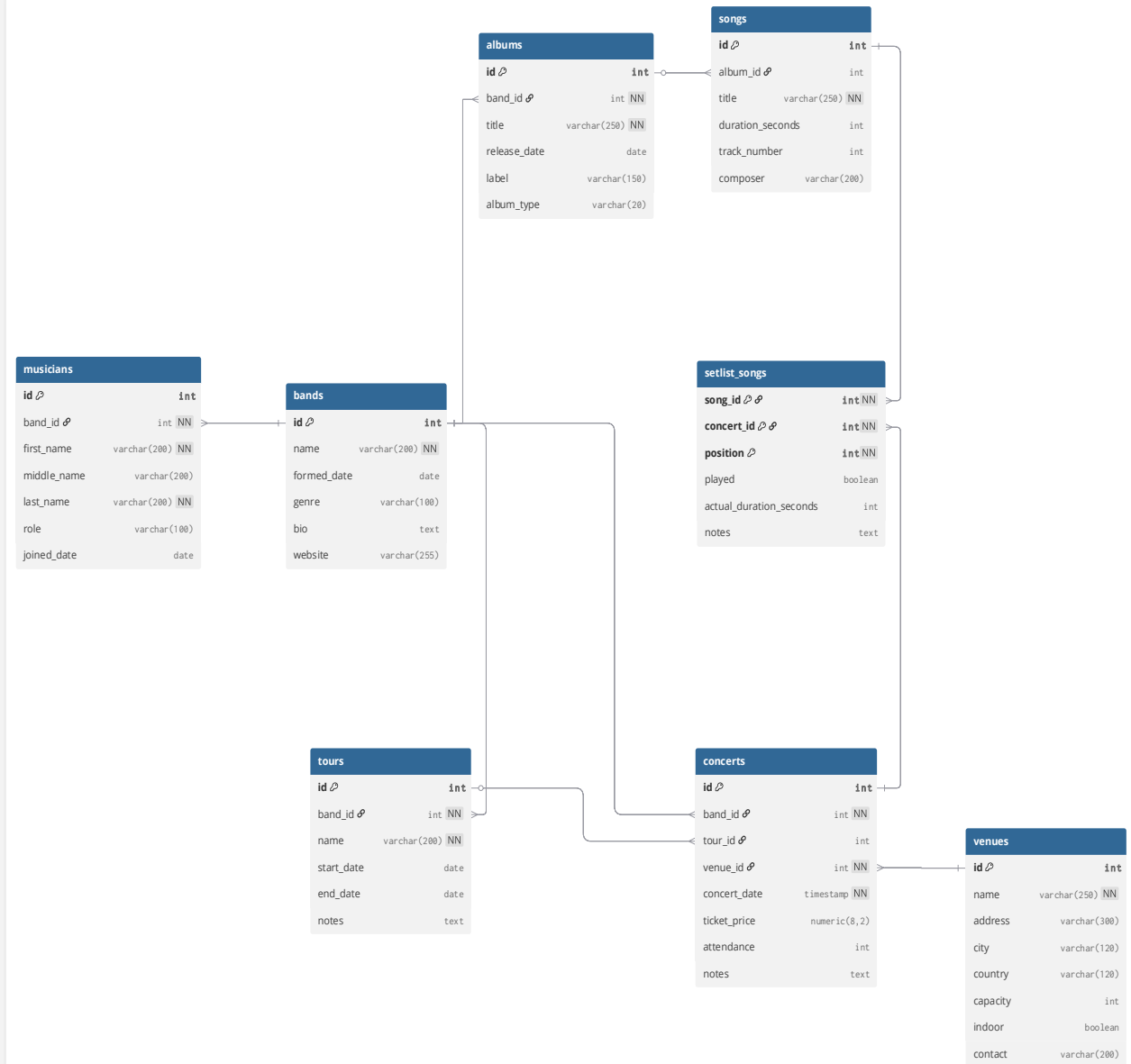
Databáze: Koncertů, Kapel, Skladeb

Adam Kijonka

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E-R diagram



Tabulky a jejich záznamy

Tabulka	Jeden záznam v tabulce reprezentuje	Obsahuje
bands	jednu hudební kapelu	název, datum založení, žánr, web, stručný popis a oficiální webovou stránku kapely.
musicians	jednoho konkrétního člena kapely	ke které kapele patří, jakou roli hraje (např. kytara, vokály), kdy se přidal
tours	Jednu konkrétní tour kapely	K jaké kapele tour patří, název tour, začátek tour, konec tour, bližší informace o tour
albums	Konkrétní Album/EP/Single kapely	K jaké kapele patří, název, datum vydání, vydavatelství a typ (Album/EP/Single)
songs	Konkrétní skladbu v Albu	K jakému albu patří, název, délku, pořadí na albu, kompozitora
venues	konkrétní koncertní místo	název, adresu, město, zemi, kapacitu, zda jde o indoor/outdoor prostor a kontakt
concerts	konkrétní koncert kapely	K jaké kapele patří, k jakému turné patří, kdy a kde se konal, návštěvnost, cenu vstupenek a doplňující poznámky
setlist_songs	skladbu zahranou (nebo plánovanou) v setlistu jednoho koncertu	pořadí, skutečnou délku provedení a volitelně poznámky

DDL – tvorba tabulek

```
CREATE TABLE [bands] (  
    [id] int PRIMARY KEY IDENTITY(1, 1),  
    [name] varchar(200) UNIQUE NOT NULL,  
    [formed_date] date,  
    [genre] varchar(100),  
    [bio] text,  
    [website] varchar(255)  
)  
GO
```

```
CREATE TABLE [musicians] (  
    [id] int PRIMARY KEY IDENTITY(1, 1),  
    [band_id] int NOT NULL,  
    [first_name] varchar(200) NOT NULL,  
    [middle_name] varchar(200),  
    [last_name] varchar(200) NOT NULL,  
    [role] varchar(100),  
    [joined_date] date  
)  
GO
```

```
CREATE TABLE [albums] (  
    [id] int PRIMARY KEY IDENTITY(1, 1),  
    [band_id] int NOT NULL,  
    [title] varchar(250) NOT NULL,  
    [release_date] date,  
    [label] varchar(150),  
    [album_type] varchar(20) DEFAULT 'Album'  
)  
GO
```

```
CREATE TABLE [songs] (  
    [id] int PRIMARY KEY IDENTITY(1, 1),  
    [album_id] int,  
    [title] varchar(250) NOT NULL,  
    [duration_seconds] int,  
    [track_number] int,  
    [composer] varchar(200)  
)  
GO
```

```
CREATE TABLE [tours] (  
    [id] int PRIMARY KEY IDENTITY(1, 1),  
    [band_id] int NOT NULL,  
    [name] varchar(200) NOT NULL,  
    [start_date] date,  
    [end_date] date,  
    [notes] text  
)  
GO
```

```
CREATE TABLE [venues] (  
    [id] int PRIMARY KEY IDENTITY(1, 1),  
    [name] varchar(250) NOT NULL,  
    [address] varchar(300),  
    [city] varchar(120),  
    [country] varchar(120),  
    [capacity] int,  
    [indoor] bit DEFAULT (1), -- TRUE  
    [contact] varchar(200)  
)  
GO
```

```
CREATE TABLE [concerts] (  
    [id] int PRIMARY KEY IDENTITY(1, 1),  
    [band_id] int NOT NULL,  
    [tour_id] int,  
    [venue_id] int NOT NULL,  
    [concert_date] datetime2 NOT NULL  
    DEFAULT GETDATE(),  
    [ticket_price] numeric(8,2),  
    [attendance] int,  
    [notes] text  
)  
GO
```

```
CREATE TABLE [setlist_songs] (  
    [song_id] int NOT NULL,  
    [concert_id] int NOT NULL,  
    [position] int NOT NULL,  
    [played] bit DEFAULT (1), --TRUE  
    [actual_duration_seconds] int,  
    [notes] text,  
    PRIMARY KEY ([song_id],  
    [concert_id], [position])  
)  
GO
```



DDL – spojení tabulek

```
ALTER TABLE [musicians]
ADD CONSTRAINT FK_musicians_band_id_bands
FOREIGN KEY ([band_id]) REFERENCES [bands] ([id]);
GO
```

```
ALTER TABLE [albums]
ADD CONSTRAINT FK_albums_band_id_bands
FOREIGN KEY ([band_id]) REFERENCES [bands] ([id]);
GO
```

```
ALTER TABLE [songs]
ADD CONSTRAINT FK_songs_album_id_albums
FOREIGN KEY ([album_id]) REFERENCES [albums] ([id]);
GO
```

```
ALTER TABLE [tours]
ADD CONSTRAINT FK_tours_band_id_bands
FOREIGN KEY ([band_id]) REFERENCES [bands] ([id]);
GO
```

```
ALTER TABLE [concerts]
ADD CONSTRAINT FK_concerts_band_id_bands
FOREIGN KEY ([band_id]) REFERENCES [bands] ([id]);
GO
```

GO

```
ALTER TABLE [concerts]
ADD CONSTRAINT FK_concerts_tour_id_tours
FOREIGN KEY ([tour_id]) REFERENCES [tours] ([id]);
GO
```

```
ALTER TABLE [concerts]
ADD CONSTRAINT FK_concerts_venue_id_venues
FOREIGN KEY ([venue_id]) REFERENCES [venues] ([id]);
GO
```

```
ALTER TABLE [setlist_songs]
ADD CONSTRAINT FK_setlist_songs_song_id_songs
FOREIGN KEY ([song_id]) REFERENCES [songs] ([id]);
GO
```

```
ALTER TABLE [setlist_songs]
ADD CONSTRAINT FK_setlist_songs_concert_id_concerts
FOREIGN KEY ([concert_id]) REFERENCES [concerts] ([id]);
```



DDL- smazání spojení a tabulek

-- DROP FOREIGN KEYS

```
ALTER TABLE setlist_songs DROP CONSTRAINT FK_setlist_songs_song_id_songs;  
ALTER TABLE setlist_songs DROP CONSTRAINT FK_setlist_songs_concert_id_concerts;  
  
ALTER TABLE concerts DROP CONSTRAINT FK_concerts_band_id_bands;  
ALTER TABLE concerts DROP CONSTRAINT FK_concerts_tour_id_tours;  
ALTER TABLE concerts DROP CONSTRAINT FK_concerts_venue_id_venues;  
  
ALTER TABLE tours DROP CONSTRAINT FK_tours_band_id_bands;  
  
ALTER TABLE songs DROP CONSTRAINT FK_songs_album_id_albums;  
  
ALTER TABLE albums DROP CONSTRAINT FK_albums_band_id_bands;  
  
ALTER TABLE musicians DROP CONSTRAINT FK_musicians_band_id_bands;  
GO
```

-- DROP TABLES

```
DROP TABLE IF EXISTS setlist_songs;  
DROP TABLE IF EXISTS concerts;  
DROP TABLE IF EXISTS venues;  
DROP TABLE IF EXISTS tours;  
DROP TABLE IF EXISTS songs;  
DROP TABLE IF EXISTS albums;  
DROP TABLE IF EXISTS musicians;  
DROP TABLE IF EXISTS bands;  
GO
```

DML – bands

```
-----  
-- BANDS  
-----  
INSERT INTO bands (name, formed_date, genre, bio, website)  
VALUES  
( 'Radiohead', '1985-01-01', 'Alternative Rock', 'English rock band from Abingdon, Oxfordshire.',  
  'https://www.radiohead.com'),  
( 'Queen', '1970-01-01', 'Rock', 'Legendary British rock band formed in London.',  
  'https://www.queenonline.com'),  
( 'Zrní', '2001-01-01', 'Alternative Folk', 'Czech alternative folk band from Kladno.',  
  'https://www.zrni.cz'),  
( 'Kabát', '1983-01-01', 'Hard Rock', 'Czech rock band from Teplice.', 'https://www.kabat.cz');
```



DML – musicians

-- RADIOHEAD

```
INSERT INTO musicians (band_id, first_name, middle_name, last_name, role, joined_date)
VALUES
(1, 'Thom', NULL, 'Yorke', 'Vocals, Guitar', '1985-01-01'),
(1, 'Jonny', NULL, 'Greenwood', 'Lead Guitar, Keys', '1985-01-01'),
(1, 'Ed', NULL, 'OBrien', 'Guitar', '1985-01-01'),
(1, 'Colin', NULL, 'Greenwood', 'Bass', '1985-01-01'),
(1, 'Phil', NULL, 'Selway', 'Drums', '1985-01-01');
```

-- QUEEN

```
INSERT INTO musicians (band_id, first_name, middle_name, last_name, role, joined_date)
VALUES
(2, 'Freddie', NULL, 'Mercury', 'Vocals, Piano', '1970-01-01'),
(2, 'Brian', NULL, 'May', 'Guitar', '1970-01-01'),
(2, 'Roger', NULL, 'Taylor', 'Drums', '1970-01-01'),
(2, 'John', NULL, 'Deacon', 'Bass', '1971-01-01');
```

-- ZRNI

```
INSERT INTO musicians (band_id, first_name, middle_name, last_name, role, joined_date)
VALUES
(3, 'Jan', NULL, 'Uhlik', 'Vocals, Guitar', '2001-01-01'),
(3, 'Ondrej', NULL, 'Skoch', 'Violin', '2001-01-01'),
(3, 'Jan', NULL, 'Juklik', 'Drums', '2001-01-01'),
(3, 'Tomáš', NULL, 'Němec', 'Bass', '2001-01-01'),
(3, 'Filip', NULL, 'Zatloukal', 'Guitar', '2001-01-01');
```

-- KABAT

```
INSERT INTO musicians (band_id, first_name, middle_name, last_name, role, joined_date)
VALUES
(4, 'Josef', NULL, 'Vojtek', 'Vocals', '1988-01-01'),
(4, 'Tomáš', NULL, 'Krulich', 'Guitar', '1983-01-01'),
(4, 'Milan', NULL, 'Špalek', 'Bass', '1983-01-01'),
(4, 'Ota', NULL, 'Vojtek', 'Guitar', '1988-01-01'),
(4, 'Radek', NULL, 'Hron', 'Drums', '1990-01-01');
```



DML- albums

-- RADIOHEAD

```
INSERT INTO albums (band_id, title, release_date, label, album_type)
VALUES
(1, 'OK Computer', '1997-05-21', 'Parlophone', 'Album'),
(1, 'Kid A', '2000-10-02', 'Parlophone', 'Album');
```

-- QUEEN

```
INSERT INTO albums (band_id, title, release_date, label, album_type)
VALUES
(2, 'A Night at the Opera', '1975-11-21', 'EMI', 'Album'),
(2, 'The Works', '1984-02-27', 'EMI', 'Album');
```

-- ZRNI

```
INSERT INTO albums (band_id, title, release_date, label, album_type)
VALUES
(3, 'Soundtrack ke konci světa', '2012-01-01', 'Indies Scope', 'Album'),
(3, 'Jiskřící', '2017-01-01', 'Indies Scope', 'Album');
```

-- KABÁT

```
INSERT INTO albums (band_id, title, release_date, label, album_type)
VALUES
(4, 'Čert na koze jel', '1991-01-01', 'Monitor', 'Album'),
(4, 'Dole v dole', '2003-10-10', 'EMI', 'Album');
```



DML - songs

-- RADIOHEAD

```
INSERT INTO songs (album_id, title, duration_seconds, track_number, composer)
VALUES
(1, 'Airbag', 280, 1, 'Radiohead'),
(1, 'Paranoid Android', 387, 2, 'Radiohead'),
(1, 'Karma Police', 260, 6, 'Radiohead');
```

-- QUEEN

```
INSERT INTO songs (album_id, title, duration_seconds, track_number, composer)
VALUES
(3, 'Death on Two Legs', 225, 1, 'Freddie Mercury'),
(3, 'Youre My Best Friend', 170, 4, 'John Deacon'),
(3, 'Bohemian Rhapsody', 354, 11, 'Freddie Mercury');
```

-- ZRNI

```
INSERT INTO songs (album_id, title, duration_seconds, track_number, composer)
VALUES
(5, 'Hýkal', 240, 1, 'Zrní'),
(6, 'Jiskřící raketa toto', 270, 2, 'Zrní');
```

--KABÁT

```
INSERT INTO songs (album_id, title, duration_seconds, track_number, composer)
VALUES
(7, 'Porcelánový prasata', 215, 1, 'Kabát'),
(7, 'Dole v dole', 250, 2, 'Kabát');
```



DML - tours

-- TOURS

```
INSERT INTO tours (band_id, name, start_date, end_date, notes)
VALUES
(1, 'OK Computer Tour', '1997-06-01', '1998-08-01', 'World tour'),
(2, 'The Works Tour', '1984-02-01', '1985-05-15', 'Worldwide tour'),
(4, 'Corrida Tour', '2006-01-01', '2007-12-01', 'Czech Tour');
```

-- VENUES

```
INSERT INTO venues (name, address, city, country, capacity, indoor, contact)
VALUES
('O2 Arena', 'Českomoravská 2345/17', 'Praha', 'Czech Republic', 18000, 1, 'info@o2arena.cz'),
('Wembley Stadium', 'Wembley', 'London', 'United Kingdom', 90000, 0, 'contact@wembley.co.uk'),
('Klub Futurum', 'Zborovská 7', 'Praha', 'Czech Republic', 800, 1, 'info@futurum.cz');
```

-- CONCERTS

```
INSERT INTO concerts (band_id, tour_id, venue_id, concert_date, ticket_price, attendance, notes)
VALUES
(1, 1, 2, '1997-09-15', 45.00, 65000, 'Radiohead at Wembley'),
(2, 2, 2, '1985-07-13', 50.00, 72000, 'Queen at Live Aid'),
(4, 3, 1, '2007-05-12', 30.00, 17000, 'Kabát Corrida Tour');
```



DML - setlists

-- Radiohead - Paranoid Android at Wembley

```
INSERT INTO setlist_songs (song_id, concert_id, position, played, actual_duration_seconds, notes)
VALUES
(2, 1, 1, 1, 395, 'Extended outro');
```

-- Queen - Bohemian Rhapsody at Live Aid 1985

```
INSERT INTO setlist_songs (song_id, concert_id, position, played, actual_duration_seconds, notes)
VALUES
(6, 2, 3, 1, 360, 'Iconic performance');
```

-- Kabát - Dole v dole at O2 Arena

```
INSERT INTO setlist_songs (song_id, concert_id, position, played, actual_duration_seconds, notes)
VALUES
(10, 3, 1, 1, 260, 'Opening song');
```



Děkuji za
pozornost

